IPv6 provides built-in auto-configuration and enhanced support for mobility and security. IPv6 also provides a much larger network address space allowing more devices to be transparently interconnected. IPv6 is gaining acceptance in the CE, mobile, and PC device industries as the long-term solution to the shortage of IPv4 addresses-while maintaining end-to-end transparency.

Support of IPv4 is essential for interoperability of devices on the digital home network, but in the longer term, IPv6 support will become more important. The future transition from IPv4 to IPv6 will be handled in the digital home design guidelines in a manner that enables devices based either on IPv4 or IPv6 to work well together.

Device and service discovery and control enables devices on the home network to automatically self- configure networking properties such as an IP address, discover the presence of other devices on the network and their capabilities, and control and collaborate with these devices in a uniform and consistent manner. The UPnP Device Control Protocol Framework (DCP Framework), Version 1, addresses all of these needs to simplify device networking in the home, and is the device discovery and control solution for digital home devices.

The UPnP Forum steering committee is currently looking at an improved version of the UPnP DCP Framework, Version 2, that integrates better with the emerging Web services model. However, for the next several years Version 1 of the UPnP DCP Framework meets the needs of the digital home and any migration to Version 2 will be handled in the digital home design guidelines in a manner that enables devices based on either Version 1 or Version 2 to work well together.

Media format and transport model

The digital home media format model is intended to achieve a baseline for network interoperability while encouraging continued innovation in media codec technology. Improvements in media codec technology result in better network bandwidth utilisation and media quality for a given bit rate. digital home requirements on media format support apply to media content that passes over the